From URDF to Current Control: A Framework for Safety-Constrained Autonomous Commissioning of Robot Manipulators

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Abstract—Commissioning a geared manipulator for direct teaching requires more than a gravity law: an engineer must verify actuator polarity, choose collision-free and informative motions, determine which parameters are observable, calibrate current-to-torque behavior, validate the result, and switch control modes safely. This paper formulates a description-driven framework designed to automate these steps from a URDF and a minimal actuator-capability description. The kinematic structure, limits, and collision geometry are trusted, whereas link dynamics and raw-current scaling need not be. The framework constructs a safe configuration set, verifies hardware sign mappings, selects experiments by information gain, identifies either a current-domain compensation map or physical gravity base parameters after independent torque-scale calibration, performs bidirectional friction sweeps, and enables current control only after held-out validation. We instantiate the current-domain identification and deployment stages on a six-degree-of-freedom DYNAMIXEL arm in ROS 2. A 181-s physical all-axis run yields chronological held-out raw-current R^2 values of 0.869, 0.952, and 0.956 for the three gravity-supported axes. Preliminary signed-current commissioning verifies the command path and exposes why a retained position loop remains stiff for hand guiding. These results establish feasibility of the sensing, fitting, and deployment path; autonomous safe-space generation, calibrated physical identification, and cross-morphology trials remain to be completed.

Index Terms—autonomous commissioning, robot dynamics identification, experiment design, gravity compensation, current control, direct teaching

I. Introduction

Direct teaching allows an operator to program a manipulator by physically guiding it. Low-cost geared arms are attractive leader devices, but they are normally commissioned through a robot-specific sequence of manual steps: check encoder and motor signs, select a safe pose, tune an excitation, fit a model, decide which joints to compensate, and finally change from a stiff position servo to a torque-like control mode. An error in any step can make the arm difficult to move or, more seriously, command gravity current with the wrong sign.

Existing gravity and friction identification methods solve important pieces of this problem [1], [2], and mature dynamic-identification work already provides observability analysis and informative trajectory design. The unresolved systems question considered here is how to connect those pieces into a validation-gated commissioning procedure that begins with a robot description and actuator feedback rather than a hand-tuned experiment.

We assume that the URDF kinematic tree, joint axes, limits, and collision geometry are available, but its inertial properties are missing or inaccurate. A hardware adapter exposes position and current-control capabilities, encoder state, raw present current, and actuator limits. Thus, the method is URDF-guided, not model-free. Its desired user interface is a URDF plus a hardware configuration; joint order, excitation, identifiable parameter set, and validation decision are generated by the system.

For slowly guided motion, the actuator torque is approximated by

$$\tau_m \simeq g(q) + r(\dot{q}), \tag{1}$$

where q and $\dot{q}$ are joint position and velocity, $g(q)$ is gravity torque, and $r(\dot{q})$ is drivetrain friction. This approximation neglects inertia, centrifugal, and Coriolis terms; it is appropriate only for the low-speed direct-teaching task considered here.

The key distinction is between predicting compensation current and recovering physical parameters. Current feedback alone can identify a useful current-domain map, but it cannot in general separate motor torque scale from mass and center-of-mass parameters. A known calibration load or other independent torque reference is required for physical identification. The framework makes this ambiguity explicit rather than labeling raw current as joint torque.

The contributions of the present framework and preliminary study are:

1) A formal commissioning pipeline that combines URDF-derived safety constraints, paired actuator-polarity verification, observability analysis, information-driven experiment selection, held-out validation, and safe controller activation.
2) A two-level identification formulation: per-joint current-domain compensation when actuator scale is unknown, and observable physical gravity parameters when an independent calibration load anchors that scale.
3) An observability-guided subtree schedule and a separate bidirectional velocity-sweep stage for friction, replacing a fixed distal-to-proximal rule and preventing exact mass/CoM claims when only base-parameter combinations are observable.

4) A preliminary OM6DOF instantiation demonstrating the current logging, chronological held-out fitting, and safety-limited signed-current deployment stages, together with a protocol for completing autonomy and scalability experiments.

II. Related Work and Positioning

Gravity compensation may be learned without a geometric model [3], while direct zero-moment teaching with self-measured gravity and friction has already been demonstrated in current control [1], [2]. Hence, neither the quasi-static law nor signed-current hand guiding is claimed as new. Recent zero-force drag work also combines minimum gravity parameters, nonlinear friction identification, current feedback, and start-up compensation [4]. Nonlinear disturbance modeling on the dVRK further shows why direction, workspace, and payload validation are necessary [5]; full dynamic identification targets a broader regime than the slow teaching task considered here [6].

Base-parameter computation and optimal robot excitation are established topics [7], [8]. Backward sequential identification has also explicitly used a distal-to-proximal structure [9], so distal scheduling is not a novelty claim. FIGAROH is particularly close to the broader direction of this paper: it constructs identification models from robot descriptions and supports observability, constrained experiment design, estimation, and validation [10]. AURT similarly automates important dynamics-calibration steps [11]. Provably safe online identification and virtual-constraint excitation design provide stronger safety treatments than the monitored constraints presently proposed here [12], [13].

Table I therefore locates the remaining integration target: hardware qualification, explicit separation of raw-current and physical parameters, staged low-speed gravity/friction experiments, and a validation- gated switch into direct-teaching current control. The present results cover only a preliminary subset of that target.

III. Commissioning Problem Formulation

For an n-joint robot, the actuator-side dynamics are

$$\tau_m = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) + \tau_f(\dot{q}) + \tau_{\text{ext}}. \tag{2}$$

Direct-teaching commissioning uses contact-free, low-acceleration experiments, so $\tau_m \simeq g(q) + \tau_f(\dot{q})$. Contact by any link other than the fixed base would add $J_c^T F_c$ and invalidate this separation; a mechanical support may therefore be used for activation and polarity checks, but not during model measurements.

The raw current observation $u \in \mathbb{R}^n$ is modeled as

$$u = A\tau_m + b + \eta, \quad A = \text{diag}(s_1 a_1, \ldots, s_n a_n), \tag{3}$$

where $a_i > 0$ is an assembled-joint raw-current/torque magnitude, $s_i \in \{-1,+1\}$ is hardware polarity, b is bias, and η includes servo and sensing error. If A and the link dynamics are both unknown, current and encoder data do not uniquely determine either one; for example, $AY_g\pi_g = (cA)Y_g(\pi_g/c)$ for any $c > 0$. The commissioner consequently supports two outputs: (i) a current-domain map $\hat{u}_g(q) + \hat{u}_f(\dot{q})$ sufficient for bounded compensation, and (ii) a physical gravity model only when an independent load or torque reference anchors A.

Inputs are the URDF topology, transforms, limits, and collision geometry; an actuator adapter declaring control modes, feedback fields, and current limits; and a minimal world description containing the base mounting surface. URDF inertial values are treated as optional priors, not ground truth.

IV. Proposed Autonomous Commissioning Framework

A. Safety Set and Hardware Verification

With joint-limit margin m_q, self/environment clearance $d_{\min}$, and a known base mounting plane, the geometric safe set is

$$\begin{aligned}\mathcal{Q}_{\text{safe}} = \{q \mid q^- + m_q \le q \le q^+ - m_q,\\ d_{\text{self}}(q), d_{\text{env}}(q) \ge d_{\min}\}.\end{aligned} \tag{4}$$

Candidate paths must also satisfy velocity, acceleration, current, temperature, and tracking-error limits. Because current demand is initially unknown, the last constraints cannot be guaranteed from geometry alone. Execution therefore begins with conservative envelopes and expands them only while online monitors remain below abort thresholds.

The URDF specifies positive joint direction, but the mapping from signed current to encoder direction may be wrong in the hardware configuration. For a supported joint, paired pulses estimate

$$s_i = \text{sgn}\left(\frac{\Delta q_i(+u_\epsilon) - \Delta q_i(-u_\epsilon)}{2u_\epsilon}\right). \tag{5}$$

Pulse magnitude is increased only until displacement exceeds an encoder threshold or a conservative current cap is reached. Pairing reduces first-order gravity drift; displacement, time, and communication watchdogs remain active. This verifies actuator polarity, not the URDF joint axis.

B. Observability and Experiment Selection

Gravity is linear in link mass and first moments,

$$g(q) = Y_g(q)\pi_g, \tag{6}$$

where each link contributes $[m_i, m_i c_{ix}, m_i c_{iy}, m_i c_{iz}]$. Stacking candidate measurements gives $\mathcal{Y}_g$. Its singular-value decomposition $\mathcal{Y}_g = U\Sigma V^T$ defines numerical rank r and the observable base parameters $\beta_g = V_r^T \pi_g$; components in the null space are not reported as individual mass or center of mass.

Before computing numerical rank, the regressor is noise-whitened and its columns are normalized; otherwise mixed mass and first-moment units make singular values and

TABLE I
Positioning of the proposed commissioning framework.

Line of work	Established capability	Boundary addressed here
Current direct teaching [1], [2], [4]	Gravity/friction current identification and zero-force control.	Commissioning raw-current actuators from a description and guarding mode activation.
Dynamic identification [7], [8], [9]	Base parameters, informative excitation, and sequential subtree estimation.	Capability-aware choice of which experiment is both informative and deployable.
Calibration toolchains [10], [11]	Description-driven model construction, constrained excitation, fitting, and validation.	Current/physical ambiguity, polarity checks, staged friction, and direct-teaching handoff.
Safe/constrained ID [12], [13]	Constrained excitation and formal safe online identification.	Conservative hardware monitors and explicit evidence gates; no formal safety claim yet.
This paper	Complete framework formulation; preliminary current-domain OM6DOF instance.	Autonomous experiment generation, physical calibration, and cross-robot evidence remain open.

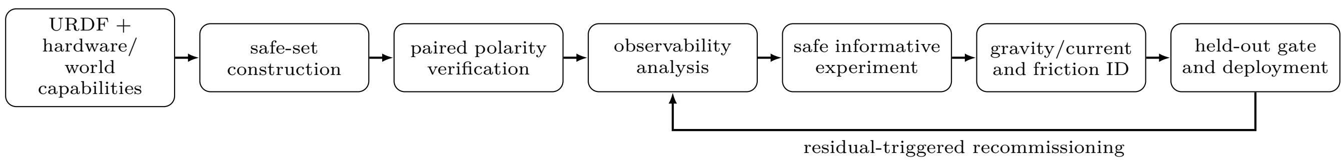

Fig. 1. Proposed autonomous commissioning loop. Each transition is guarded by geometric, actuator, data-quality, and model-validation checks.

condition numbers arbitrary. If gravity, friction, and bias are estimated together, observability is evaluated on the augmented regressor, not $\mathcal{Y}_g$ alone.

The next safe experiment e is selected from candidate poses or trajectories by an incremental information–risk objective,

$$e^* = \arg\max_{e\in\mathcal{E}_{\text{safe}}} \left[\log \frac{\det(F_{\text{prev}} + F_e + \lambda I)}{\det(F_{\text{prev}} + \lambda I)} - \lambda_R R(e)\right], \tag{7}$$

where F_e is the normalized regressor information matrix and $R(e)$ penalizes clearance, current, and tracking risk. A Fourier, multisine, or spline path is only the trajectory parameterization; its amplitudes, frequencies, and joint participation are outputs of (7).

For joint i, gravity contains contributions from its descendant subtree,

$$\tau_{g,i} = \sum_{\ell\in\mathcal{D}(i)} Y_{i\ell}(q)\pi_\ell. \tag{8}$$

A calibrated, observable distal subtree can be subtracted before estimating a parent only in physical mode, where all joints share consistent torque units. Per-joint current-domain coefficients cannot be propagated as link parameters. Moreover, a distal joint may have zero gravity sensitivity or expose only one parameter combination. We therefore use observability-guided subtree scheduling: distal-first is preferred when it increases rank at low risk, but it is not imposed as a universal identification rule.

C. Current Scale, Gravity, and Friction

Without an independent torque reference, each joint is fitted directly in the command domain,

$$u_i = \phi_{g,i}(q)^T\theta_{g,i} + u_{f,i}(\dot{q}_i) + b_i, \tag{9}$$

where $\phi_{g,i}$ contains the nonzero, SVD-reduced kinematic columns of the ith gravity-regressor row. Thus, $\theta_{g,i}$ is described only as an effective current-domain parameter. For physical identification, a payload of measured mass and center of mass is attached at a known transform. Matched loaded and unloaded measurements give [14]

$$\Delta u = A\Delta g_p(q), \qquad \Delta g_p(q) = \frac{\partial V_p(q)}{\partial q}, \tag{10}$$

which estimates the observable entries of A. Axes receiving negligible payload torque require a different load orientation or an external torque reference. With A anchored, robust least squares estimates only the base parameters supported by the SVD analysis.

Friction is identified after gravity using constant-speed passes through the same configurations in both directions, after trimming acceleration segments. Outside a stiction dead zone,

$$\tau_{f,i}(v) = \begin{cases} f_{c,i}^+ + f_{v,i}^+ v, & v > v_\epsilon, \\ -f_{c,i}^- + f_{v,i}^- v, & v < -v_\epsilon. \end{cases} \tag{11}$$

with nonnegative $f_c^\pm$ and $f_v^\pm$. At matched position and speed magnitude, the half-sum of positive/negative measurements emphasizes gravity, while their half-difference emphasizes odd friction. This staged experiment is preferred to interpreting a quantized multisine velocity trace as a final friction measurement.

D. Validation-Gated Deployment

The model is tested on unseen configurations, velocity directions, and a separate payload condition. The gate checks current-prediction error, released pose drift, saturation frequency, temperature, polarity, and fault response. Physical transparency additionally requires a handle-force measurement. Let γ_i be one only when polarity, rank, confidence interval, held-out error, Mode 0 transfer, and actuator/fault tests pass. Deployment also requires the uncertainty envelope

$$|\mu_{u,i}(q)| + k\sigma_{u,i}(q) \leq u_{\text{safe},i} \quad \forall q \in \mathcal{Q}_{\text{deploy}}. \tag{12}$$

The command applies $\Gamma = \text{diag}(\gamma_i)$ before saturation and slew limiting. A failed gate blocks free Mode 0 activation and requires a mechanically supported state or safe handoff to position control; zero current alone is not safe for an unsupported gravity-loaded arm. Residual-triggered recommissioning is evaluated only during declared contact-free self-checks or tool-change events, because human teaching force is itself a residual.

V. Preliminary OM6DOF Instantiation

A. Platform and Existing Excitation

The preliminary platform is a six-DOF vertical OM6DOF arm driven by XM430-series DYNAMIXEL actuators [15] through ROS 2 Humble and ros2_control [16]. Its effort state is raw Present Current, not measured joint torque. The deployed controller evaluates a KDL gravity prior and folds gripper/camera branches into the nearest chain link. The configured 0.114-kg distal mass includes an unmeasured 0.08-kg camera/bracket estimate. Offline analysis instead finite-differences the potential energy of the expanded URDF tree. Numerical equivalence of these two gravity implementations remains to be verified.

The existing teaching_all data are a manually configured precursor, not output of the complete autonomous experiment designer. Each joint followed three sinusoidal terms,

$$q_i(t) = q_{i,0} + \sum_{k=1}^{3} A_{ik} \sin(\omega_k t), \tag{13}$$

at 0.5, 0.8, and 1.2 rad/s, within manually checked joint-limit margins. The run contains 18,156 samples over 181.05 s at 100.27 Hz. It was executed under the position trajectory controller; transferring its effective current model to signed Current Control is therefore an explicit assumption.

Figure 2 shows the three axes retained for preliminary gravity support. Reported velocity is quantized in approximately 0.02397-rad/s increments. T_g^{URDF} is a nominal model prediction, and Present Current is a raw register value.

B. Chronological Current-Domain Fit

For this precursor data set, the nominal gravity feature is fitted as

$$u_i = c_{g,i} g_i^{\text{URDF}}(q) + c_{c,i} \tanh(\dot{q}_i/\epsilon) + c_{v,i}\dot{q}_i + b_i. \tag{14}$$

Here c_g is raw current per nominal-model Nm: it combines actuator scale and URDF inertial error and is not a physical torque constant. Candidate friction terms are used only for prediction and remain disabled in deployment.

TABLE II
Chronological held-out raw-current prediction on the existing OM6DOF run.

Joint	g^{URDF} [Nm]	c_g [raw/nom. Nm]	RMSE	R^2
J2	$[-0.282,\ 0.219]$	176.39	10.63	0.869
J3	$[-0.699,\ -0.404]$	165.54	9.67	0.952
J5	$[-0.283,\ -0.211]$	139.34	5.89	0.956

Every tenth row is retained (1,816 samples); samples with $|\dot{q}_i| < 0.005$ rad/s and the first/final 2 s are omitted. Training uses $[2, 120)$ s, followed by a 2-s guard interval, and testing uses $[122, 179.3)$ s. This is an unseen time block of the same trajectory, not workspace or payload generalization.

The other test R^2 values are 0.383 (J1), 0.526 (J4), and 0.263 (J6). Selecting J2/J3/J5 was a conservative manual decision, not an automatic observability gate. Figure 3 shows the prediction. The quantized velocity and position-loop data preclude interpreting c_v as a validated physical friction coefficient.

VI. Preliminary Signed-Current Deployment

A. Current-Based Position Control Baseline

DYNAMIXEL Operating Mode 5 retains an internal position loop [15]. In this mode, Goal Current acts as a current ceiling rather than a signed torque command. Experiments with a position setpoint plus a current limit produced a stiff, non-smooth response. Reducing the position deadband made the arm sag, whereas increasing it made the arm difficult to move. This trade-off is inherent to the remaining position loop; on the present platform it did not provide transparent hand guiding. A quantitative force comparison remains part of the final evaluation.

B. Pure Current Control

The prototype teaching profile uses Operating Mode 0. The commanded current is

$$u_i^* = \text{sat}_{u_{\max,i}} \left(\hat{u}_{g,i}(q) + \hat{u}_{f,i}(\dot{q}) + \hat{b}_i \right), \tag{15}$$

where sat applies a symmetric current limit. The initial physical trial uses $\hat{u}_{f,i} = 0$ and $\hat{b}_i = 0$, so that the gravity path and activation behavior are validated before uncertain friction terms are introduced. Full gravity–friction compensation is enabled only after bidirectional constant-speed validation shows that these residual terms improve held-out predictions and release drift. A 0.5-s current ramp and conservative per-joint caps limit activation transients.

Unlike Mode 5, zero current is neutral in Mode 0. Consequently, the controller does not hold a pose by itself; a supported arm, an active controller, current limits, and a communication fail-safe are required. The normal position-control stack and the teaching stack are mutually exclusive hardware owners.

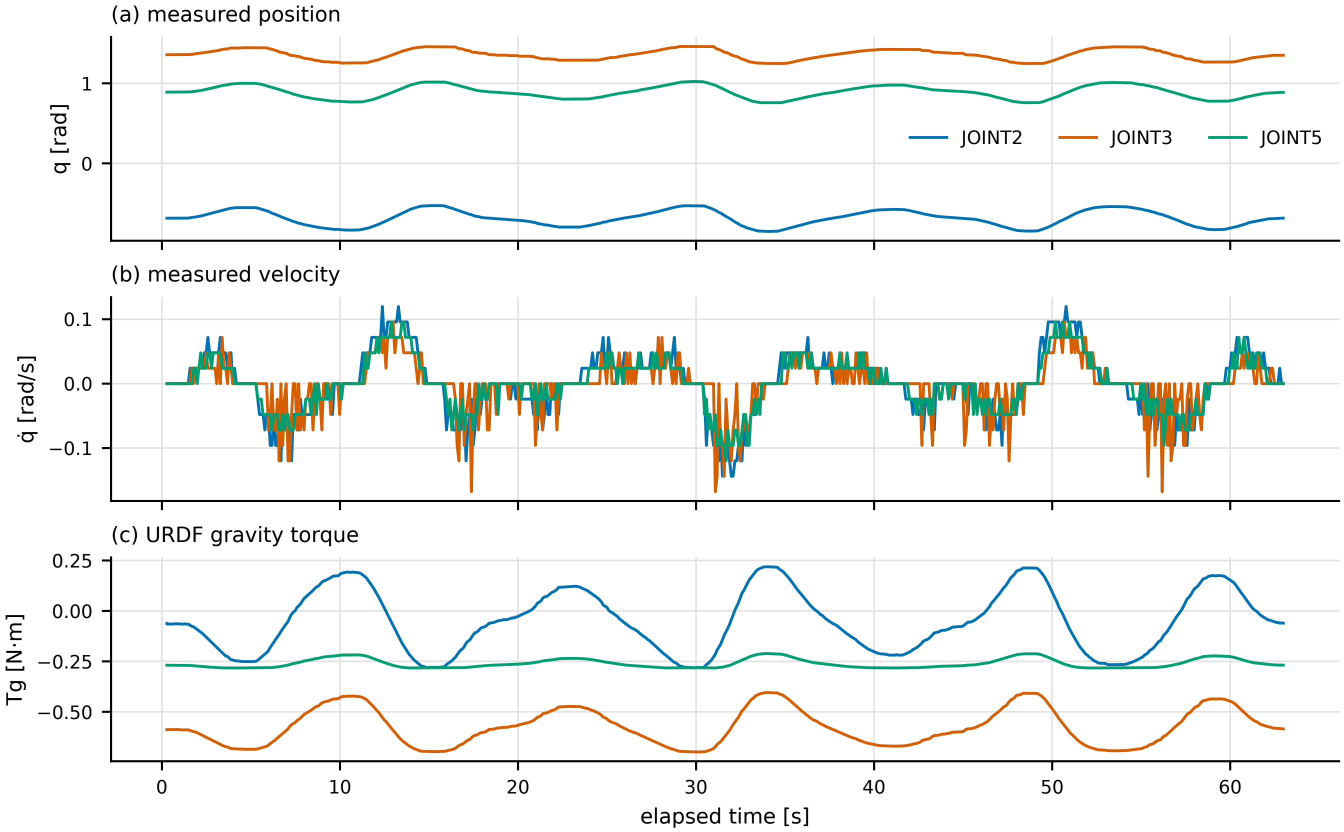

Fig. 2. Existing OM6DOF multisine data over the first 63 s. Position and reported velocity are measured; T_g^{URDF} is the whole-tree model prediction. This is preliminary manually bounded excitation, not yet an autonomously synthesized experiment.

VII. Preliminary Commissioning Results

At a representative supported pose, the gravity model produced $g_2(q) = 0.1090$ Nm for joint 2. With the identified scale, the controller commanded 19.38 raw current ticks, while the measured present current was 18 ticks. This agreement verifies raw-current command tracking at that pose; it does not verify physical gravity torque or identify actuator torque scale. The arm was qualitatively more backdrivable than with the Mode 5 baseline. A small residual joint-2 drift remained, indicating that bias, friction, gravity-model error, or Mode 5-to-Mode 0 transfer error is still present.

These observations are gravity-only commissioning evidence, not a complete gravity–friction performance evaluation. No interaction force, payload sweep, temperature sweep, or repeatability statistic has yet been measured. The current results therefore support feasibility of the deployment path but do not establish general superiority over other hand-guiding methods or a validated friction-compensation benefit. The Mode 0 spot check predates the chronological reanalysis in Table II; it is not an independent validation of the held-out fitted parameters.

VIII. Evaluation Plan

The existing experiment evaluates only current-domain prediction and a single-pose deployment path. Table III defines the measurements needed to evaluate the complete framework. Unmeasured entries are protocols, not claimed results. Each physical condition will be repeated at least three times, with timestamped position, velocity, Goal Current, Present Current, voltage, temperature, controller state, and hardware-fault status.

RQ1 counts every user choice after the URDF, hardware adapter, and mounting surface are supplied, including ready pose, joint order, parameter subset, amplitude, and frequency. RQ2 first calibrates A with a measured payload at matched poses, then evaluates on a second payload and unseen workspace poses. The optimized trajectory is compared with the existing manual multisine and a safe random baseline using normalized regressors.

RQ3 records minimum collision distance in simulation and hardware current, tracking, and watchdog aborts during bounded fault injection. RQ4 applies the same software pipeline to modular 2-, 3-, 4-, and 6-DOF descriptions, then adds a payload and measures residual detection and recovery after recommissioning. RQ5 uses a calibrated handle load cell; subjective ease of motion is supplemen-

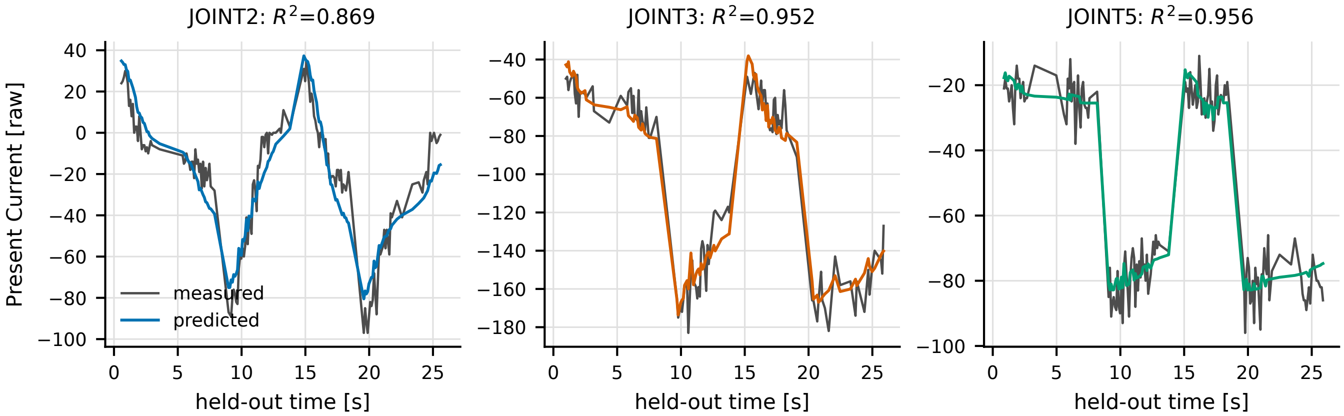

Fig. 3. Held-out Present Current prediction. Parameters use only $[2, 120)$ s; displayed test samples begin at 122 s. Both measured and predicted quantities are raw current, not torque.

TABLE III
Evaluation matrix. Only the preliminary portion of RQ2 is currently reported in Table II.

Question	Comparison	Primary metrics
RQ1 autonomy	manual vs. generated commissioning	interventions; completion; time
RQ2 information	manual, random-safe, and optimized excitation	rank; $\sigma_{\min}$; condition; held-out error
RQ3 safety	nominal and injected faults	clearance; peak current; abort latency
RQ4 transfer	2/3/4/6-DOF and changed payload	success rate; residual; recovery
RQ5 teaching	Mode 5, nominal Mode 0, calibrated Mode 0	handle force; drift; jerk; saturation

tary only. Release drift and the RMS/95th-percentile jerk use identical filters across controllers. Confidence intervals are computed by trial-level bootstrap rather than treating adjacent time samples as independent.

IX. Safety and Reproducibility

Changing DYNAMIXEL operating mode resets actuator settings; the procedure must disable torque, write and verify zero Goal Current, start only one hardware owner, and then enable the teaching controller with the arm supported. The controller must command zero current upon deactivation or invalid state data. Identification begins only after the support is removed and collision checks confirm that no distal link contacts the mounting surface. Per-axis current and slew caps, state timeouts, temperature limits, a dead-man input, and a physical emergency stop remain outside the learned model.

For each submitted result, the expanded URDF, controller configuration, firmware/model/ID register dump, supply voltage, payload, code commit, and raw log will be archived. The candidate friction terms identified from multisine data will be validated through bidirectional constant-speed sweeps independently from the multisine run before they are enabled. The offline whole-tree and deployed KDL gravity implementations must also agree numerically on the logged configuration set before a physical-parameter result is reported.

X. Limitations and Threats to Validity

The full autonomous commissioner is a proposed framework; the current implementation covers logging, predefined excitation, current-domain fitting, and safety-limited deployment. It does not yet generate collision-free experiments, perform payload-based torque calibration, or automatically gate axes by observability. The manually selected multisine was collected under an internal position servo, and velocity is quantized; transfer to Mode 0 and the quasi-static assumption require dedicated validation.

Exact link masses and centers of mass are generally not individually observable, and raw current cannot resolve their scale without an independent load. Friction can vary with direction, temperature, supply voltage, and gearbox history. The configured camera/bracket mass is unmeasured. Finally, cross-morphology scalability and automatic recommissioning are research hypotheses until RQ1–RQ4 are completed.

XI. Conclusion

This paper formulates commissioning as the complete transition from a robot description to validated current control, rather than as gravity fitting alone. The framework combines safe-set construction, actuator-polarity verification, observable-parameter analysis, information-driven experiment selection, separate gravity/friction experiments, and a deployment gate. It also separates current-domain compensation from physical identification so raw actuator feedback is not mislabeled as torque. Existing OM6DOF data support feasibility of the logging, held-out fitting, and signed-current command path, but not yet the autonomy, physical accuracy, scalability, or friction benefit of the full framework. The evaluation plan turns those gaps into explicit experiments for the next revision.

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